

E-Commerce Conversion Funnel & Revenue Analysis

Where customers drop off, which categories drive revenue, and what a typical customer is worth.

67M

ROWS ANALYZED

22.36%

VIEW → CART RATE

\$623

AVG. CUSTOMER LTV

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1. Results

VIEW → CART

22.36%

CART → PURCHASE

53.45%

VIEW → PURCHASE

11.95%

BOTTLENECK

22.36% **WEAK SPOT**

View → Cart conversion, the biggest leak in the funnel

OVERALL CONVERSION

11.95%

View → Purchase across the full funnel

TOP CATEGORY

Electronics

\$205.2M in revenue, 7× the next-best category

AVERAGE CUSTOMER LTV

\$623.12

Average total spend per unique purchasing customer

2. Project Preview

Main Question

What is the drop-off rate between events? Calculate the conversion rate at each funnel step (view → cart → purchase).

Secondary Questions

- What are the top 5 best and worst main categories by revenue?
- What is the average customer LTV?

Dataset Overview

- **Shape:** 67M rows, 9 columns
- **Type of analysis:** Conversion Funnel & Abandonment Analysis

Schema

COLUMN	DESCRIPTION
<code>event_time</code>	When the event occurred
<code>event_type</code>	User action: view, cart, or purchase
<code>product_id</code>	Unique ID for a product listing
<code>category_id</code>	Unique ID for a category listing
<code>category_code</code>	Compounded category string, e.g. "electronics.smartphone"
<code>brand</code>	Product brand
<code>price</code>	Product price at time of event
<code>user_id</code>	Unique ID for a shopper
<code>user_session</code>	Browsing session the action belongs to

Main Dataset Problems

1. Data types

- `event_time` is a string, not a datetime
- `product_id`, `user_id` are Int64, allocating more space than needed
- `price` is Float64, allocating more space than needed

2. Duplicates and unusable nulls

- `user_session` has 10 null rows
- ~0.14% of rows are exact duplicates

3. Categorization

- `category_code` is compounded (e.g. "electronics.smartphone") and contains up to 4 levels of depth (e.g. "electronics.audio.music_tools.piano")

4. Missing values

- `category_code`: 32.44% null
- `brand`: 13.66% null

3. Application

Fixing Dataset Problems

1. Data types

- `event_time`: string → datetime
- `product_id`, `user_id`: Int64 → Int32
- `price`: Float64 → Float32

2. Duplicates and unusable nulls

- Dropped all rows where `user_session` is null
- Duplicates were most likely caused by page reloads (the same `user_id` triggering repeated identical events). To catch true duplicates rather than normal repeated user activity, rows were compared on their entire content, including `event_time`, rather than on a single ID column.

3. Categorization

- `category_code` has up to 4 levels of depth, but full granularity wasn't needed for this analysis. Split into two columns instead: `category` and `sub_category`.
- Example: "electronics.audio.music_tools.piano" → category = "electronics", sub_category = "audio.music_tools.piano"

4. Missing values

- `category` / `sub_category`: nulls (32.44%) replaced with "Unknown Category" / "Unknown Sub Category"
- `brand`: nulls (13.66%) replaced with "Unknown Brand"
- `user_session`: 10 null rows deleted

Sinking the Cleaned Data

Since the dataset exceeds 67M records, the cleaning and standardization steps were run once and streamed to a new output file. This avoids re-running the full cleaning pipeline every time the data needs to be analyzed.

Answering the Main Question

1. Conversion rate at each step (view → cart → purchase)

Grouped by `event_type` and counted the number of unique `user_id` s at each step, then computed:

STEP	FORMULA	RATE
View → Cart	$(\text{cart} / \text{view}) \times 100$	22.36%
Cart → Purchase	$(\text{purchase} / \text{cart}) \times 100$	53.45%
View → Purchase (overall)	$(\text{purchase} / \text{view}) \times 100$	11.95%

2. Top 5 best/worst categories by revenue

Filtered to purchase events only, grouped by category, summed price per group, then sorted descending.

Top 5 Best (by revenue):

MAIN CATEGORY	TOTAL REVENUE	TOTAL ORDERS
electronics	\$205,249,741.53	493,637
<i>unknown_category</i>	<i>\$29,879,190.27</i>	<i>234,214</i>
appliances	\$18,640,385.14	99,023
computers	\$13,994,330.68	34,477
furniture	\$2,543,251.02	11,542

Top 5 Worst (by revenue):

MAIN CATEGORY	TOTAL REVENUE	TOTAL ORDERS
sport	\$395,692.75	1,489
accessories	\$98,203.02	2,154
medicine	\$16,442.95	344
country_yard	\$12,447.13	60
stationery	\$3,212.48	191

3. Average customer LTV

Summed price for every purchase per unique `user_id` , then averaged those totals across all users.

Average LTV = \$623.12

4. Conclusion & Recommendations

Specific Notes

1. Fix the browsing experience, not checkout

The cart → purchase conversion rate is healthy, but there's a large gap between view → cart. Effort should go toward improving the browsing experience (better product images and descriptions on view pages) rather than the checkout process, since that's where the majority of users are lost.

2. Protect and prioritize Electronics

"Electronics" generates 7× the revenue of the next-best category. This category should be prioritized and protected, since it disproportionately drives the business's overall revenue.

General Notes

1. Join with marketing data

Join this data with marketing data to calculate metrics like Customer Acquisition Cost (CAC). This would meaningfully sharpen future decision-making.

2. Review-for-discount system

Add a review-for-discount system (e.g. a 5% discount for an honest product review) to improve the browsing experience and give potential customers an added incentive to purchase.

3. Loyalty points system

Add a loyalty points system for customers who leave honest reviews after a purchase, covering both the product and the overall buying experience. Points could be redeemed for discounts or offers. This would systematically increase retention and customer LTV, while also lowering the cost of acquiring new customers by building public trust and confidence.